

# **RAJALAKSHMI ENGINEERING COLLEGE**

## **(Autonomous)**

**RAJALAKSHMI NAGAR, THANDALAM, CHENNAI-602105**



**AI23632 - Natural Language Processing**

**LABORATORY RECORD NOTEBOOK**

**Name : GOKUL PRASATH P**

**Year / Branch / Section : III- AIML/A**

**Register No : 2116231501051**

**Semester : VI**

**Academic Year : 2025-2026**

# RAJALAKSHMI ENGINEERING COLLEGE

[AUTONOMOUS]

RAJALAKSHMI NAGAR, THANDALAM – 602 105

## **BONAFIDE CERTIFICATE**

Name : GOKUL PRASATH P

Academic Year : 2025-2026

Semester : 06

Branch : AIML

Register No.

2116231501051

Certified that this is the bonafide record of work done by the above student in the  
**AI23632 – Natural Language Processing** during the year 2025 - 2026.

Signature of the Faculty In- charge

Submitted for the Practical Examination held on .....

Internal Examiner

External Examiner

# RAJALAKSHMI ENGINEERING COLLEGE

(An Autonomous Institution affiliated to Anna University)

## DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

### AI23632 - Natural Language Processing

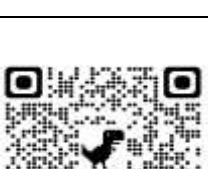
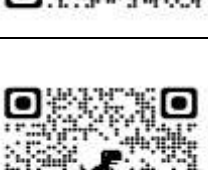
Reg No: 2116231501051

Name: GOKUL PRASATH P

Department : AIML

#### INDEX

| S.NO | Category Title                                                                                                   | QR Code                                                                              | Signature |
|------|------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|-----------|
| 1.   | <b>To preprocess text using NLTK techniques</b><br>Including tokenization, cleaning, stemming, and lemmatization |    |           |
| 2.   | <b>To compute entropy, cross entropy, and perplexity</b><br>For evaluating language model performance            |   |           |
| 3.   | <b>To perform rule-based morphological analysis</b><br>Identifying prefixes, roots, and suffixes in words        |  |           |
| 4.   | <b>To implement morphological analysis using FSA</b><br>Using state transitions for structured word segmentation |  |           |
| 5.   | <b>To build a unigram language model</b><br>Applying Laplace smoothing for probability estimation                |  |           |
| 6.   | <b>To implement a bigram language model</b><br>Using multiple smoothing techniques for better predictions        |  |           |

|     |                                                                                                              |                                                                                      |  |
|-----|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--|
| 7.  | <b>To develop a spelling correction system</b><br>Using edit distance and bigram probabilities               |    |  |
| 8.  | <b>To compute TF-IDF and cosine similarity</b><br>And evaluate word associations using PMI                   |    |  |
| 9.  | <b>To visualize word embeddings using PCA and t-SNE</b><br>And explore dimensionality reduction techniques   |    |  |
| 10. | <b>To implement Hidden Markov Model algorithms</b><br>Using forward, backward, and Viterbi methods           |   |  |
| 11. | <b>To perform word sense disambiguation and parsing</b><br>Using Lesk algorithm, dependency, and CYK parsing |  |  |
| 12. | <b>To implement Named Entity Recognition using HMM</b><br>Applying Viterbi decoding for sequence labeling    |  |  |